

Finetuning LLMs for Grammatical Error Correction in English and Greek Texts

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Abstract

Accurate grammatical correction is paramount for effective communication, especially for non-native speakers of a language. This research aims to harness the power of the generative capabilities of LLMs, such as the T5 model and transfer learning to develop an efficient and flexible system for automated grammatical correction in written text. It involves the finetuning of a pre-trained T5 model on a custom dataset containing English sentences with varying degrees of grammatical errors. Data preprocessing involves encoding the sentences in the T5 format, allowing it to generate corrected sentences for input with grammatical errors. The results show the effectiveness of the improved T5 model in grammatical correction. We also report results for a low-resourced scenario of Greek texts, using the multilingual model's variant. The model achieves competitive performance on benchmarking metrics, outperforming existing methods in terms of accuracy and contextual understanding. The findings highlight the importance of using pre-trained models and finetuning techniques to develop sophisticated grammar correction systems and writing aids.

1 Introduction

Understanding the peculiarities of grammar and syntax remains a challenge, both for native and non-native speakers of a language. Grammatical errors in written texts can impede comprehension and reduce the overall quality of content. Recognizing the widespread presence of these errors and their impact on communication, the field of grammar error correction (GEC) has evolved over time. Numerous approaches to grammar

correction have been explored ranging from rule-based systems and statistical models to machine translation (Ailani et al., 2019), for example, GEC applications found in word processors, web-based writing services, such as Grammarly, etc.

Traditional methods, although useful to some extent, often struggle to address the specificities and context-dependent nature of grammatical errors. In response to these, the field of Natural Language Processing (NLP) has undergone a transformation with the advent of neural network-based models, offering the promise of more accurate and context-aware grammatical correction (Madi and Al-Khalifa, 2018). The two shared tasks, CoNLL-2014 (Ng et al., 2014) and BEA-2019 (Bryant, Felice, Andersen, & Briscoe, 2019), demonstrate the increasing attractiveness of GEC as a Natural Language Processing (NLP) topic.

In this paper we aim to harness the power of the generative capabilities of the T5 (Text-to-Text Transfer Transformer) model (Shazeer et al., 2019) to develop an efficient and flexible system for automated grammar correction in written text. The proposed methodology involves finetuning a pre-trained T5 model on a customized dataset containing sentences with varying degrees of grammatical errors. The model employs the sequence-to-sequence function, allowing it to produce corrected sentences for input with grammatical errors. In addition, we investigate its performance on a low-resources scenario involving custom made datasets of Greek texts, thus accounting for the morphological complexities of mid- to low-resourced languages, such as Greek. It is shown that finetuning can perform well even in the absence (or total lack thereof) any specific language pretraining.

Experimental results highlight the effectiveness of the fine-tuned T5 model in grammatical correction. We also comparatively evaluate the

model against other current LLMs and ensemble approaches and show it achieves competitive performance on evaluation metrics, outperforming existing methods in terms of accuracy and contextual understanding. The present study provides insights into the strengths of the model, its ability to handle diverse error patterns, and areas for further improvement.

To our knowledge, there is no systematic review and evaluation of finetuning language models for the specific task of grammar correction, while experiments have been carried out on a low-resourced Greek dataset using both T5 and its multilingual variant, mT5 (Xue et al., 2020) and evaluated with the GLEU metric (Napoles et al., 2015). Our source code is openly available at: <https://github.com/DennisKappelles/Grammatical-error-correction-in-JFLEG-dataset-with-T5-text-to-text-transformer>.

The rest of this paper is organized as follows: in Section 2 we review relevant literature in the field of grammatical error correction; in Section 3 we present our methodology and approach, by outlining the error correction procedure designed with the T5 model and the adaptation to the Greek language. Section 4 summarizes the evaluation configuration, key metrics and loss function used. Section 5 contains the results of our experiments and their analysis as well as a comparative evaluation with current state-of-the-art. Finally, section 6 outlines our conclusions and future work, while limitations are discussed in the last section.

2 Related work

Several studies have shown that GEC can be approximated with machine translation using a Seq2Seq model (Luong et al., 2015) along with more recent Transformer architectures (Zhang et al., 2023). Previous work has utilized various advanced models. Transformer-based models (Junczys-Dowmunt et al., 2018), such as ours, leverage the classical Transformer architecture (Vaswani et al., 2017), which excels at sequence-based tasks due to its self-attention mechanisms that capture long-term dependencies within sentences. Another approach combines Statistical Machine Translation (SMT) with a BiGRU network (Grundkiewicz and Junczys-Dowmunt, 2018), enabling the model to understand context from both directions of a sequence, enhancing its corrective capabilities. Copy-augmented model (Wei Zhao, 2019) introduce a "copy mechanism"

that allows for directly copying segments from the input to the output, which is particularly useful in GEC since many words and phrases remain unchanged in corrections. Additionally, Transformer models pre-trained with pseudo data and enhanced with BERT-based model (Masahiro Kaneko, 2020) improve language understanding and the quality of generated corrections. Tagged corruption model (Felix Stahlberg and Shankar Kumar, 2021) introduce intentional "errors" in the training data, helping the model to recognize and correct standardized mistakes within sentences. VERNet (Zhenghao Liu and Tat-Seng Chua, 2021) is a specialized model for GEC that uses tailored approaches to handle complex language structures and grammatical errors more effectively. Finally, CNN Seq2Seq model (Shamil Chollampatt1 and Hwee Tou Ng, 2018) leverage Convolutional Neural Networks for sequence-to-sequence transformations, with Seq2Seq architecture used to learn the relationships between original and corrected sentences.

The T5 model has been effectively applied in several grammatical error correction tasks and related language processing projects. For example, in GECToR (Omelianchuk et al., 2020), T5 was used as a reference model to compare the effectiveness of tagging versus text-to-text correction for error correction. (Kaneko et al. 2020) applied T5 for correcting mistakes made by second-language learners, demonstrating T5's adaptability to varied language structures. In the BEA 2019 Shared Task, T5 was utilized by various research teams for GEC, leveraging its text-to-text structure for accurate error correction. Finally, (Kiyono et al. 2019) combined T5 with data augmentation techniques to enhance GEC performance, particularly for low-resource datasets. These projects showcase T5's flexibility and efficiency in addressing grammatical correction and other complex linguistic tasks.

To account for the morphological complexities of the Greek language, a related effort is presented (Katerina Korre, John Pavlopoulos, 2022), which introduced methods to enhance GEC resources specifically for Greek. This study used datasets built from Greek learner corpora and evaluated the models using metrics such as Precision, Recall, and F0.5. Datasets used include the Greek Native Corpus (GNC) and the Greek Learner Corpus Corrections (GLC2). The GNC includes essays by Greek high school students, annotated for errors

and corrections, while the GLC2 contains corrections by Greek philologists on essays by learners of Greek as a second language. The best-performing model, a Transformer-based architecture, achieved an F0.5 score of approximately 60%, indicating strong performance in prioritizing precision over recall in GEC tasks for Greek.

ELERRANT, an automated tool for classifying grammatical error types in Greek was developed and presented (Katerina Korre, Marita Chatzipanagiotou, John Pavlopoulos, 2021). ELERRANT was trained on an annotated corpus of Greek learner texts with error types categorized according to linguistic taxonomy. Evaluated with accuracy and F1-score metrics, the system achieved an F1-score of around 70% on error-type classification, showing strong potential for accurately identifying and categorizing grammatical errors in Greek.

3 Methodology

3.1 Dataset

A JFLEG dataset is used to train the model. JFLEG (JHU Fluency-Extended GUG) is a dataset for grammatical error correction (GEC) in English (Courtney Napoles, Keisuke Sakaguchi, Joel Tetreault, 2017). It is a gold reference standard for developing and evaluating GEC systems in terms of fluency (whether a text resembles its native language) as well as grammar. For each original text (sentence), there are four corrections written by observers (corrections) following specific guidelines.

Table 1. A sample of the JFLEG dataset with data.

sentence	corrections
I want to talk about nocive or bad products like alcohol , hair spray and cigarrets .	["I want to talk about nocive or bad products like alcohol , hair spray and cigarettes . ", "I want to talk about harmful or bad products like alcohol , hair spray and cigarettes . ", "I want to talk about harmful or bad products like alcohol , hair spray and cigarettes . ", "I want to talk about harmful or bad products like alcohol , hair spray and cigarettes . "]
For not use car .	["Not for use with a car . ", "Do not use in the car . ", "Car not for use . ", "Can not use the car . "]

There are several reason .	["There are several reasons . ", "There are several reasons . ", "There are several reasons . ", "There are several reasons . "]
Thus even today sex is considered as the least important topic in many parts of India .	["Thus , even today , sex is considered as the least important topic in may parts of India . ", "Thus , even today , sex is considered the least important topic in many parts of India . ", "Thus , even today , sex is considered the least important topic in many parts of India . ", "Thus , even today sex is considered as the least important topic in many parts of India . "]
For example they can play football whenever they want but the olders can not .	["For example , they can play football whenever they want , but the elders cannot . ", "For example , they can play football whenever they want but the others can not . ", "For example , they can play football whenever they want but the seniors ca n't . ", "For example they can play football whenever they want but the older ones cannot . "]

Each instance contains a source sentence and four corrections. Sentence field contains the original sentence written by an English learner and corrections field contains corrected versions by human annotators. The order of the annotations are consistent (eg first sentence will always be written by annotator "ref0"). The dataset contains a total of 1,503 records and is divided into two roughly equal parts validation (755 rows) and test (748 rows). The dataset is available on the Hugging Face dataset provider network and can be accessed using the Datasets library.

3.2 Models Used

The T5 model, developed by Google, is a transformer-based language model designed to handle a wide range of natural language processing (NLP) tasks in a unified "text-to-text" format. This means that it takes both input and output as text strings, making it adaptable to tasks such as translation, summarization, question answering, and grammatical error correction. T5 was trained on the large-scale C4 dataset and is notable for its flexibility, enabling fine-tuning for diverse NLP applications while maintaining strong performance across tasks. In this task we use the pre-trained transformer model *t5-base*, the variant of the T5 model, which is considered the basic model for

such treatments and has about 220 million parameters.

A multilingual variant of T5, mT5 (multilingual T5) (Linting Xue, 2020), can be used for Greek datasets, as it has been pre-trained on data from many languages, including Greek. mT5 was pre-trained on a dataset covering 101 languages, following the paradigm of the more general solution of pre-training on multiple languages. Greek ranks 20th in data representation, with 43 billion tokens, making mT5 well-suited for Greek NLP tasks. The least common languages were Sotho and Yoruba, and other indigenous languages that have a much smaller amount of data available. In our task, we use one of the variants of the Mt5 model, Mt5-base, which contains 580 million parameters.

Both T5 and mT5 use an Encoder Decoder transformer which is pre-trained with masked language modeling, covering successive intervals of input tokens and then attempting to reconstruct them. T5 was pre-trained on 750 GB of English-language text derived from the public web Common Crawl. mT5 was pre-trained on data from all 71 monthly web data published by Common Crawl so far, which is more than the source data used by T5.

3.3 Implementation

Our work is carried out in the Google Colab environment. This environment comes with 16GB memory and uses GPU (T4 GPU) for hardware acceleration, which helps due to the large amount of data that needs to be generated for the task. We will use a Python package, Happy Transformer, to implement our work. Happy Transformer, built on top of Hugging Face's Transformers library, facilitates fine-tuning and inference with NLP Transformer models, as well as implementing and training transformer models.

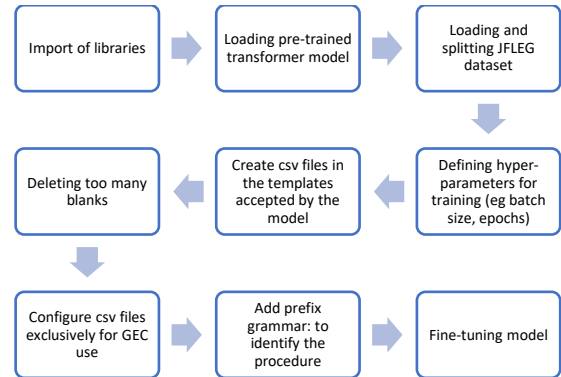
The process begins with loading the pre-trained model t5-base. Before we proceed with training the model, we need to preprocess our dataset, to correct any errors it contains and improve the performance of the grammar correction. Some of the instances within the dataset contain too many blanks and if not corrected, the model will produce blanks when not required.

The next part of the preprocessing is to bring the dataset into the appropriate format for Happy Transformer. We need to structure both the training data and the evaluation data in the same format,

which is a CSV file with two columns: input and target. The input column contains grammatically incorrect text and the target column contains text with the corrected version of the text from the input column. In total, we create 3,016 training examples and 2,988 evaluation examples.

The hyper-parameters defined for fine-tuning the model are: the batch size equal to 8 and the epochs equal to 10. In addition, we use a *default learning rate* equal to $5e-5$, *max_input_length* and *max_output_length* equal to None and *fp16* equal to False. *Max_input_length* and *max_output_length* reflect the maximum number of tokens for input and output. The rest are truncated. By default, the maximum number of tokens the model can handle are used. If *fp16* is true, it enables half-precision training, which saves space by using 16 bits instead of 32 to store the model weights. After having finished all the preprocessing, we are ready to fine-tune our pre-trained transformer model. A summary of the implementation workflow is presented in Fig. 2.

Figure 1. Implementation workflow.



For our task we need to specify which procedure we want to perform by adding the same prefix to each input. In this case, we will use the "grammar:" prefix. This is because T5 models are able to perform multiple tasks, such as translation and summary, with a single model and a unique prefix used for each task, so that the model knows which task to perform.

3.4 Adaptation to Greek

As with the T5 model used for the English texts, we follow the same procedure and technique for the mT5 model for the Greek texts. At the same time, we perform experiments with Greek texts against the original T5 (t5-base) to showcase ablation. To

account for the low-resourced Greek scenario, we use two manually built datasets, one for fine-tuning the model and one for evaluating it, with 50 incorrect sentences and their corresponding corrections each. These datasets follow the patterns that the mT5 model accepts as input and are sufficient for its correct operation. The dataset structure is similar to that of JFLEG. The datasets consist of two columns containing incorrect sentences with their corresponding corrections.

Inspired by ELERRANT, the key metric for this process, we have introduced specific types of errors to tailor our dataset to the specific morphology of the Greek language. For example, we have added errors of accent, as accent in Greek is indicated by a punctuation mark (') and not just by the emphasis of the word during speech. In addition, errors have been added concerning the final n. The retention or omission of the final n in certain Greek words (articles, pronouns or particles), because of its frequency as an error even by native speakers, is considered a different type of error and not simply a spelling error.

For finetuning mT5, we use mt5-base model. We use batch size and epochs as previous and the default learning rate of $3e-5$, *min_source_length* equal to 256 as the *max_target_length*, which are adequate for sentence size. Sample contents of the two datasets are presented in Table 2.

Table 2. Sample contents of the two Greek datasets (training and evaluation)

Sentence	Corrections
Εχω ένα βιβλίο .	Έχω ένα βιβλίο.
Αυτη είναι τι σπιτι μου .	Αυτή είναι το σπίτι μου.
Εμεις παμαι στο παρκο καθε μερα .	Εμείς πάμε στο πάρκο κάθε μέρα.
Αυτοι ειναι οι καλυτερη φιλη μου .	Αυτοί είναι οι καλύτεροι φίλοι μου.
Το σκυλος ειναι μεγαλο και γρηγορο.	Το σκυλί είναι μεγάλο και γρήγορο.

4 Evaluation

4.1 Metrics

The incorrect sentence originally written is referred to as the source sentence in the evaluation indicators of grammatical error correction systems. The manually marked and corrected sentence is known as the reference sentence and the sentence corrected by the grammar error correction system is known as the hypothetical sentence. The manually marked and corrected sentences are the

best standard for evaluation and are used to compare with the hypothetical sentences to evaluate the performance of the system (Manli Long, 2022).

GLEU (Napolet et al., 2015) is a variant of the BLEU metric, adapted for evaluating GEC systems. It measures the n-gram overlap between corrected proposals and reference proposals, taking into account both the accuracy and recall of the n-grams. Accuracy measures the percentage of corrections made by the GEC system that are actually correct. High accuracy indicates that the model makes few incorrect corrections. Recall measures the percentage of actual errors in the text that the GEC system detects and corrects correctly. High recall indicates that the model detects most errors. GLEU provides a balanced view of the performance of the GEC system, taking into account both the fluency and adequacy of the corrections. It is particularly useful for capturing subtle grammatical refinements that may be missed by other metrics.

GLEU usually ranges between 0.0 and 1.0. If the two sentences are perfectly matched, then $GLEU = 1.0$. Conversely, if the two sentences do not match at any point, then $GLEU = 0.0$. The core of the GLEU evaluation metric is the detection of the number of words of common occurrence between the hypothetical sentences and the reference sentences.

The mathematical formula for calculating the GLEU metric is as follows:

$$GLEU = BP * \exp \sum_{n=1}^N wn \log pn \quad (1)$$

$$pn = \frac{N(H, R) - (N(H, S) - N(H, S, R))}{N(H)} \quad (2)$$

$$BP = \begin{cases} 1, & \text{if } h > r \\ \exp(1 - \frac{r}{h}), & \text{if } h \leq r \end{cases} \quad (3)$$

In the above formulas, pn is the post n-gram accuracy and BP is the penalty factor. In (2), H is the length of the hypothetical sentence and R is the length of the reference sentence. The function of the penalty factor is to avoid the bias of the scoring system. In the scoring process, the n-gram matching score can be made better by reducing the length of the sentence. Therefore, the length of the sentence is taken into account. N is the value of n in the n-gram of GLEU type and its upper bound is 4. $N(H, R)$ are the overlapping n-grams in the hypothetical sentence and the reference sentence and $N(H, S, R)$ are the overlapping n-grams in the hypothetical sentence (Hypothetical), the source

sentence (Source) and the reference sentence (Reference), respectively. wn is the weighted average adopted by the system and its value is $1/N$.

We expect GLEU to be inversely proportional to Cross-Entropy Loss due to formula (1). As is obvious, the mathematical type of cross entropy loss is contained in the exp of equation (1) and, as will be shown in section 5, it is experimentally verified that the two quantities are in fact inversely proportional.

4.2 Loss Function

The loss function used is Cross-Entropy Loss. Cross-Entropy Loss measures the difference between two probability distributions: the true distribution (the base truth value) and the predicted distribution (the model output). For sequence-to-sequence tasks, it is calculated at each step of the sequence and averaged over all steps.

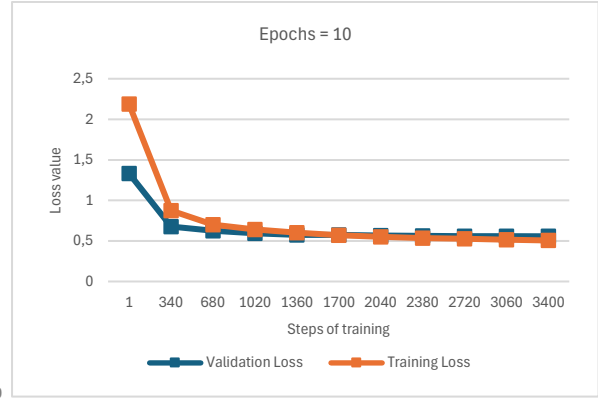
For each word in the target sequence (corrected sentence), Cross-Entropy Loss compares the predicted probability distribution in the vocabulary with the encoded actual distribution (the actual word). This loss is calculated at each time step (token) in the output sequence. Token-level losses are summed or averaged over the entire output sequence to obtain the total loss for that sequence.

The two losses of the model after each step are the training loss and the validation loss. Training loss is a method used to evaluate how a model fits the training data. This means that it evaluates the error of the model on the training set. The training set is a portion of a dataset used for initial model training. Computationally, the training loss is calculated by taking the sum of the errors for each instance in the training set. It is also important to note that the training loss is measured after each batch. This is usually illustrated by plotting a curve of training loss.

The validation loss is a method used to evaluate the performance of a model on the validation set. The validation set is a portion of the data set that is defined to validate the performance of the model.

The validation loss is similar to the training loss and is calculated from the sum of the errors for each instance in the validation set. In addition, the validation loss is measured after each iteration. This informs us whether or not the model needs further tuning or adjustments. Fig. 2 indicates the distribution of losses during finetuning.

Figure 2. Curves of training and validation losses.



5 Results and Discussion

5.1 T5 fine-tuning results on the English dataset (JFLEG)

In Table 3, we report the results of GLEU metric and loss function before and after the fine-tuning of our pre-trained model.

Table 3. Evaluation results on JFLEG.

Stage of implementation	GLEU score	Loss score
t5-base before fine-tuning	0.0945	1.28039
t5-base after fine-tuning	0.7764	0.47939

We observe that the model loss is significantly high before training. This means that the model is not yet able to correctly predict the expected outcome, i.e. the correct sentences. So, the learning problem essentially becomes an optimization problem by defining a loss function and then optimizing (training) the algorithm to minimize the loss function. As can be seen, the loss after training is significantly reduced. This means that the model is trained correctly, and we have achieved the optimization we are looking for.

We use a list of sentences and the T5 model to generate corrected sentences from the uncorrected ones. Now that we have the candidate sentences (predictions) available, we can calculate the GLEU score before fine-tuning. To compute it, we use references which is the list of reference sentences using predictions which is the list of prediction sentences (model output).

As we can see, the value of the metric is quite higher than the one before fine-tuning, which indicates that our model mostly produces correct results. Approximately 77 out of 100 suggestions produced by our model agree with the target

suggestions of the evaluation dataset. This makes our model reliable for the grammatical error correction task and ranks among the best practices.

5.2 Comparative Evaluation

Table 4. GLEU score of our task compared to scores of models applied to JFLEG dataset

Rank	Model	GLEU score
1	Our task of GEC with T5	77.6
2	Tagged corruptions	64.7
3	VERNet	62.1
4	Transformer (EncDec architecture) + Pre-train with Pseudo Data + BERT	62.0
5	SMT + BiGRU	61.5
6	Copy-augmented Model (4 Ensemble + Denoising Autoencoder)	61.0
7	Transformer (self-attention-based model)	59.9
8	CNN Seq2Seq	57.47

Table 4 shows the GLEU scores of the various models used for GEC and reviewed in related work, on JFLEG-type datasets. We observe that using a Transformer-based model such as ours, we achieve a performance of about 60% and more specifically 59.9. This practically means that our model shows higher performance than other models of the same type using Transformers. This is because in our work we also achieved a performance close to 77%, which is up to 17% more than other similar models.

Moreover, we see that using a combination of models SMT + BiGRU increases the GLEU score by about 2%. Model combinations that contain their own unique features are likely to cover or even correct imperfections that the models may exhibit individually. Grammatical error correction using a Transformer-based model ranks 7th in the GLEU score ranking table, while grammatical error correction using a Transformer + Pre-train with Pseudo Data + BERT-based model ranks 4th in the GLEU score ranking table over a JFLEG dataset. In conclusion, the Tagged corruptions model and the VERNet model achieve the highest performances compared to the previous works, but their GLEU score values are still lower than the one presented here, while the CNN Seq2Seq model achieves the lowest performance among all the models.

5.3 mT5 and T5 finetuning results on the Greek dataset

Following the same steps, we calculate the loss and the GLEU score of the mT5 model before fine-tuning, fine-tune the mT5 model on the Greek training dataset and then calculate the new loss and evaluate it using the GLEU metric using the Greek evaluation dataset. As a means of ablation, we repeat the same experiment with the original T5, which has been pretrained with Greek texts. The results obtained are shown in Table 5.

Table 5. Evaluation results of mt5 and T5 models on the Greek dataset

Model	GLEU score before fine-tuning	GLEU score after fine-tuning	Loss score before fine-tuning	Loss score after fine-tuning
Mt5	0.0715	0.4226	1.41930	1.09475
T5	0.0378	0.3106	1.73729	1.38173

Fig. 3 and 4 show the loss curves of the mt5 and T5 models during their fine-tuning back to the Greek dataset. During the training, the training and the validation losses decrease, which shows that the procedure is correctly executed.

Figure 3. Curves of training and validation losses for the mT5 model

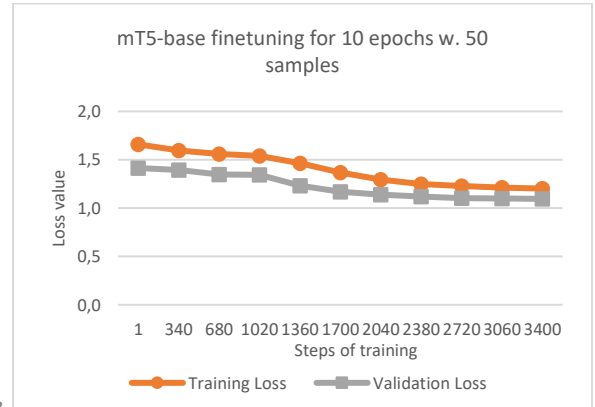
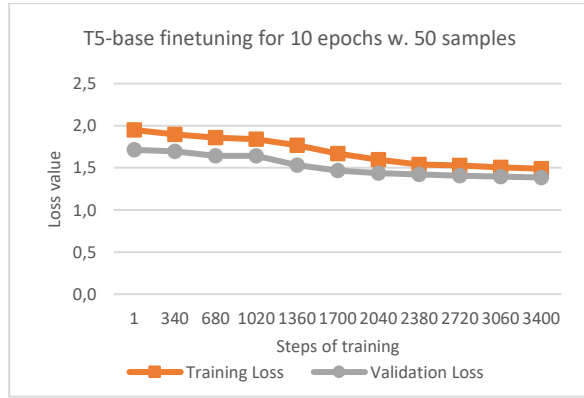


Figure 4. Curves of training and validation losses for the T5 model

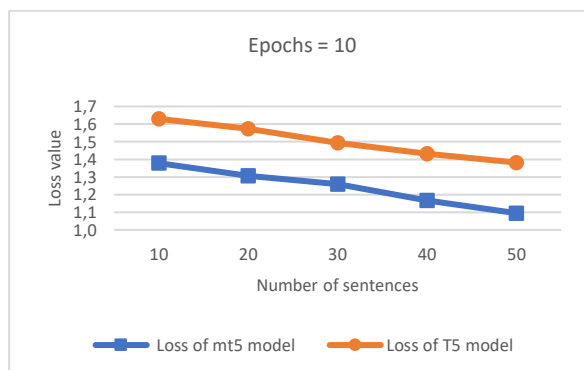


By observing the loss before and after fine-tuning, we understand that although the loss of the model is reduced, it remains at very high levels. This is due to the very small data sample we provide to the model, as on a much larger data set it would show better results. We observe the same in the GLEU score for our model. Similarly to before, the GLEU score increases after fine-tuning but remains at very low levels, around 42%. In Table 4 we observe that the mt5 model performs better compared to the T5 model, which indicates that mt5 is more suitable for Greek language tasks, due to the multilingual training it has received.

However, when low resources are the case, both in terms of available data as well as computational power, finetuning seems to pay off (Table 5): even with only 10 epochs and 50 samples, the original T5 of 220m outperforms the much more expensive 580m mT5 although pretrained on 43b Greek tokens.

Still, finetuning addresses the specificities of a domain and can only improve both models: Fig. 5 indicates that the loss of the models after fine-tuning decreases as the size of the dataset increases, which proves that the size of the dataset is important for the performance of these systems.

Figure 5. Loss value after fine-tuning with mt5 and T5 models



6 Conclusions and future work

Through the evaluation we have presented some useful conclusions. Our work shows satisfactory results in correcting grammatical errors compared to previous work, with a rate of about 77% in the GLEU benchmark. The model loss during training shows quite a significant difference, as its value before fine-tuning is 1.28 and ends up at 0.47. This is indicative of the capability of fine-tuning to adapt the model means that our model is trained correctly and produces correct results.

The score achieved by our work using the GLEU metric exceeds the score of many works in previous years using Transformer architecture and approaches the levels presented by other models (BERT, GPT, hybrid approaches).

Dataset size and quality in a low-resources scenario is also shown to be important. The volume of parallel data in GEC is not comparable to even the largest Lang-8 dataset (1,147,451 sentences). The size of the data corpus in languages other than English is significantly smaller. Worse, when applied to other less spoken languages where there is not a large amount of parallel data, the performance of many powerful MT-based GEC systems will be severely degraded. As indicated by this work, a feasible solution may come from carefully fine-tuning existing language models; whenever possible, pre-training and data augmentation strategies can help to incorporate large amount of error-free text for low-resource languages.

Limitations

Although adequate for finetuning, our work uses a Greek dataset that is relatively and purposely small. A larger dataset might be useful, especially when representing greater GEC domains (native speakers, foreign learners etc.). Especially for L1 (the author's first language), treating as equivalent texts written by authors with different first languages causes small returns on evaluation metrics (F0.5) approaching 50%.

Comparison and relation to even larger generative LLMs such as GPT, Llama, Gemini and Claude could be investigated. Few- to zero-shot learning can be leveraged to analyze and compare their performance on GEC, with a particular focus on languages they are not pre-trained with.

More specific error metrics and penalty weights need to be devised and tested for highly inflected

- languages such as Greek. In addition, metrics do not measure meaning retention, which is also necessary to consider when evaluating a GEC system.
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